Bed 1 Pump-Drive Circuit — Rebuild & Verification Procedure

ZAS 2026 — Smart Energy-Driven Irrigation for Agricultural Value Chains · generated 2026-09-09

Written after a bench test where the relay audibly clicked but the pump never ran, and multimeter checks found current leaking through the coil even at idle (pin 86→GND = 12.9V correct; pin 85→GND = 6V; 85-to-86 = 5.6V drop). That pattern means current is flowing through the coil right now, when the transistor should be fully off — almost certainly the 2N2222 or the 1N4007 landed on the wrong physical leg. This procedure verifies both components directly, in isolation, before anything gets reconnected.

> **Step 0 — Disconnect, safely**
>
> 1. Disconnect the 12V battery/bus at the source, not just at the relay. Confirm 0V between the fused bus and ground with a multimeter.
> 2. Physically unclip the 2N2222 and the 1N4007 from the circuit so each can be tested in isolation.

Step 1 — Verify the 1N4007 diode (diode-test mode, component in isolation)

1. Multimeter to diode-test mode.
2. Probe both leads, one direction then the other. One direction reads ~0.5–0.7V (forward bias: red probe = anode, black probe = cathode/banded end). The other direction reads open (OL).
3. **Write down which physical end is the cathode** — it must land on pin 86.
4. A reading in both directions = shorted, replace it. No reading in either direction = open, replace it.

Step 2 — Verify the 2N2222 transistor (diode-test mode, component in isolation)

Don't trust the datasheet's generic pin order — verify the actual part in hand.

1. Label the three TO-92 legs A, B, C arbitrarily.
2. Test all 6 probe combinations. The one leg that gives a forward-bias reading to *both* other legs when the red (+) probe is on it is the **Base**.
3. Distinguish Collector from Emitter: a slightly lower Base-to-Collector drop than Base-to-Emitter on a precise meter, or use your multimeter's hFE/transistor-tester socket if it has one — only the correct orientation reads a plausible hFE.
4. **Write down which leg is Base, Collector, and Emitter.**

Step 3 — Reconnect, one stage at a time, testing after each

3a. Base drive only. ESP32 GPIO18 → 220Ω resistor → the verified Base leg. Power the ESP32 only (USB), 12V bus still disconnected. Trigger briefly and confirm the base leg's voltage changes.

3b. Add Emitter + Collector. Emitter → common ground. Collector → relay pin 85 (wire run only, relay/coil not yet powered). Still no 12V.

3c. Add the coil circuit + diode. Pin 86 → fused +12V bus. Pin 85 → Collector. 1N4007 cathode → pin 86, anode → pin 85/Collector. Reconnect the 12V bus.

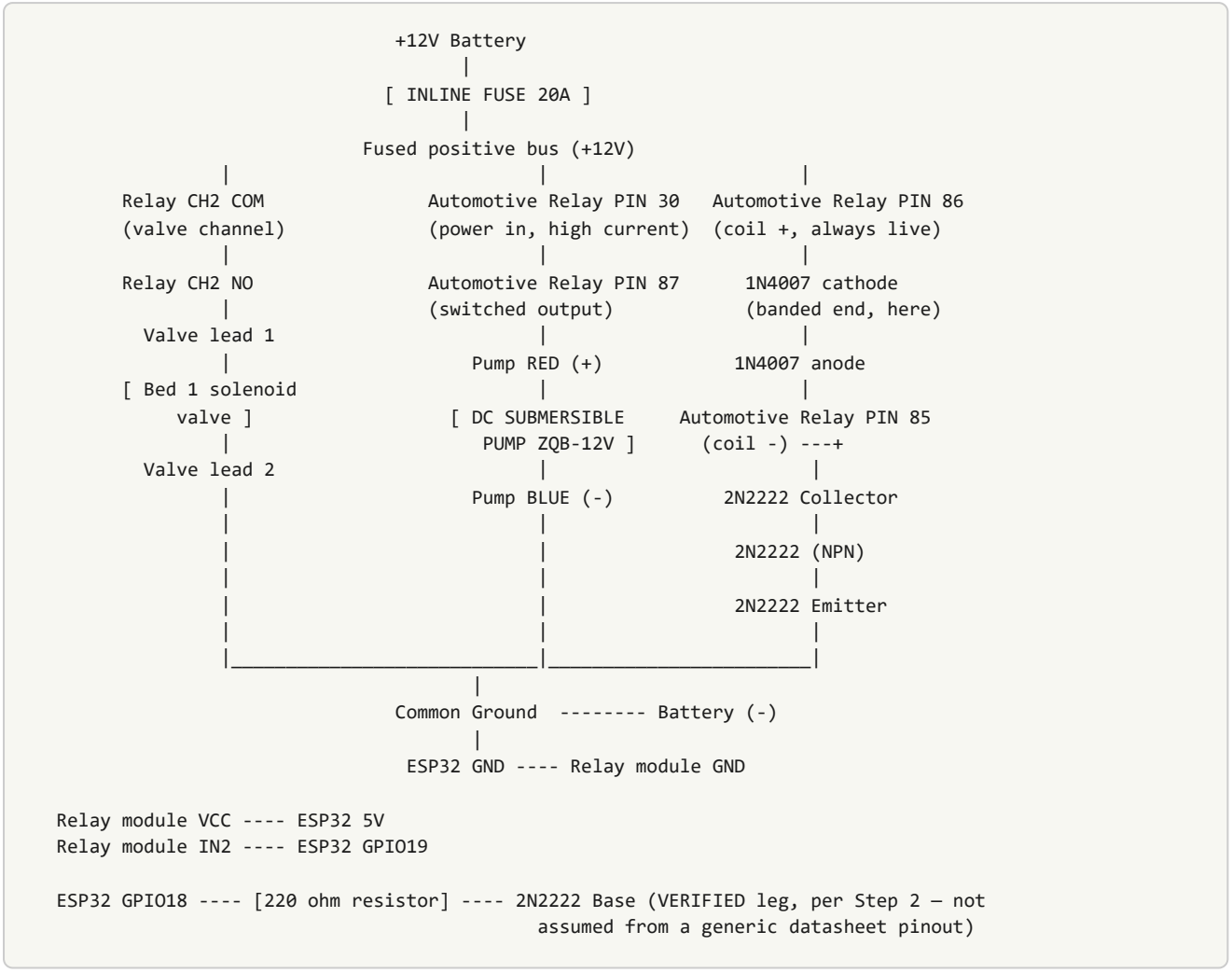
Idle test: pin 86→GND ≈ 12V; pin 85→GND should now be *close to 12V too*, not ~6V — if it's still dropping, stop and recheck orientation before continuing.
Triggered test: during RUNNING, pin 85→GND should drop close to 0V, and the relay click should be solid, not soft.

3d. Add pins 30/87 (no pump yet). Pin 30 → fused +12V bus (separate wire, same bus). Idle test: pin 30→GND ≈ 12V. Triggered test: pin 87→GND should now read close to 12V during RUNNING — this is the check that read 0V before.

3e. Add the pump (bucket of water, not the real bed). Red (+) → pin 87. Blue (−) → ground. Trigger a short (5s) cycle and confirm the pump actually runs and moves water.

3f. Add the valve. Only after 3e is solid — reconnect the relay-module channel exactly as before (GPIO19 → IN2, CH2 COM → +12V, CH2 NO → valve lead 1, valve lead 2 → ground). Test a full end-to-end cycle.

Circuit diagram (target state, after rebuild)



Why this order

Each stage only adds one new thing to the previous, already-verified state, with its own pass/fail check before moving on — so if something's still wrong, the failing stage tells you exactly which addition caused it, instead of rebuilding the whole circuit and hoping.

Source: `firmware/sensor-node/bed1-manual-irrigation/BED1-PUMP-CIRCUIT-REBUILD.md`. Companion to `bed1-manual-irrigation.ino` in the same folder.